

Do skills predict moves? A gravity test of skill similarity against observed occupational transitions

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Working paper. The model, its inputs and its readings are fixed; the estimation waits on the bilateral transition matrix.

Extended summary

A whole advisory industry now tells workers and workforce boards which occupations they can move to, and it does so with a similarity score.¹ Two occupations whose skill profiles overlap at 98.5 per cent are presented as a few months of retraining apart, a machinist is pointed at industrial maintenance, and public money follows the pointer into bridge programmes and reskilling grants. The score is a claim about feasibility: that a person with one profile could do the other job. Whether anyone makes the move is a different claim, it is the one the advice depends on, and it has not been tested at scale, although the data to test it are public and the test is cheap.

This paper specifies the test and fixes its inputs. Observed flows between occupations come from the March supplement of the Current Population Survey, which records in one interview what a person does now and what they did last year, and which the companion flow account (Conway, 2026a) has shown to be the only instrument in that survey that measures occupational transitions at all: two independent codings of a job the survey itself records as unchanged disagree 48.1 per cent of the time at the detailed level, so a matrix linked across interviews measures coding disagreement, and the disagreements fall along classification boundaries, elementary against secondary teachers, supervisors against the staff they supervise, which are exactly the pairs a similarity score ranks highest. A linked panel would therefore find similarity predicting moves whether or not anyone moved. The within-interview design cancels that error, and it is the one used here. Similarity comes from a composite over the skills, knowledge and work activities the Occupational Information Network records, held as 798,342 occupation pairs above a similarity of 0.5, on which the tier thresholds in workforce practice sit at 0.985, 0.970 and 0.950.

The model is a gravity equation estimated by Poisson pseudo-maximum likelihood on gross flows with origin and destination effects, so that the size of each occupation is absorbed and the elasticity of flows to similarity is identified from how flows vary across destinations within one origin, holding the wage differential. Similarity is symmetric in a pair and the wage differential is antisymmetric, which splits the test in two: with origin and destination effects the level of traffic between a pair identifies the similarity elasticity, and with pair effects added similarity drops out and the wage differential is identified from the direction of that traffic alone. The flow account has already shown that between a third and two fifths of gross lateral movement carries direction at the pair level and that a tenth changes the size of any occupation, so the two halves ask what a similarity score can be expected to explain, which is traffic, and what it cannot, which is where the traffic nets to. A flexible version of the model, with similarity entered in bins rather than as one elasticity, reads off where observed flows break and puts the breaks against the three thresholds in use. Licensing enters as a reading rather than a regressor: the pairs that similarity places close and the account shows nobody

crossing are the map of what the score misses, and whether that map concentrates in licensed occupations is a question the residuals answer.

1 The claim under test

Skill-similarity scores are built from occupational profiles. Each occupation in the Network carries importance ratings on thirty-five skills, thirty-three knowledge areas and forty-one work activities, and a composite similarity between two occupations is the cosine of their two rating vectors, weighted one half on skills and a quarter each on knowledge and activities. The scores are used in career tools, in workforce board planning, in the design of bridge programmes and in the shortage arithmetic of the companion paper (Conway, 2026b), where a tier of adjacency decides which occupations can release workers into a short one. In every one of those uses a high score is read as evidence that people move.

The score says something narrower. It says that the two jobs draw on the same abilities in similar proportions, which is a statement about what a worker could do, and a worker's decision to move depends on that and on a wage differential, on where the destination jobs are, on whether a licence stands in the way, on what the worker knows about the destination, and on whether the destination is hiring. Gathmann and Schönberg (2010) showed that workers do move along task-distance gradients in German administrative data, and Kambourov and Manovskii (2008) showed that occupation-specific human capital is large enough to keep people where they are. Neither result says what a point of similarity buys in realised movement in the United States at the detailed level, and neither tested the thresholds that a practitioner's tool now draws as lines between an easy move and a career change. That elasticity, and those thresholds, are what this paper estimates.

2 Why the matrix has to come from one interview

The natural object is a transition matrix: how many people were in occupation i last year and are in occupation j now. The survey has two ways of building one, and the flow account measures the difference between them. Following a person across two interviews a year apart, with occupation coded independently on each occasion, returns an annual mobility of 50.1 per cent at the detailed level; restricting to 114,491 people whom the survey itself records as holding the same occupation in both years and working all fifty-two weeks, the two codings still disagree 48.1 per cent of the time. The March supplement instead asks in one interview what a person does now and what they did last year, both codes are assigned at one sitting from one respondent's description, and the annual mobility it returns on the same linked people is 10.3 per cent. The errors of the two codes are correlated, and the correlation cancels in the difference.

For this paper the choice is decisive for a reason beyond precision. The independent-coding disagreements follow the boundaries of the classification: the largest single confusion is elementary against secondary school teachers, then supervisors against the staff they supervise, then generic against specific managerial categories. Those are pairs with near-identical skill profiles, so a matrix built from linked interviews carries spurious transitions concentrated exactly where similarity is highest, and a gravity model on it would return a large positive elasticity that measures the coding process. The within-interview matrix is the flow account's own construction, year by age band by origin by destination for the 230 occupations the survey supports, and every cell carries a standard

error from the 160 replicate weights the Census Bureau publishes. It is built and it is not yet published; this paper runs on it when it is.

3 The model

Write f_{ij} for the weighted number of people observed moving from occupation i to occupation j over the pooled years, s_{ij} for their composite similarity, and w_i and w_j for the mean wages of the two occupations in the Occupational Employment and Wage Statistics. The estimating equation is

$$\mathbb{E}[f_{ij}] = \exp\left(\eta \ln s_{ij} + \beta \ln \frac{w_j}{w_i} + \alpha_i + \delta_j\right),$$

estimated by Poisson pseudo-maximum likelihood, which keeps the pairs where nobody moves rather than dropping them and handles the heteroskedasticity a log-linear specification mishandles (Santos Silva and Tenreyro, 2006). The origin effect α_i absorbs the size of the origin, its overall propensity to lose people and everything else that moves its departures as a whole; the destination effect δ_j does the same for arrivals. What is left to identify η is the variation in similarity across the destinations of one origin, and the variation in origins of one destination, holding the wage differential between them.

The migration paper (Conway, 2026c) supplies the lesson on identification. Estimating a wage elasticity one occupation at a time in that setting identified it only from year-to-year movement in a survey cell mean, of which a reliability calculation attributed 54 per cent to sampling noise, and the per-occupation estimates were a null produced by the design. Pooling identified the elasticity from cross-sectional differences at a reliability of 0.76, and the estimate came out at 0.114, weak and precisely measured. The same holds here: the pooled model identifies η from cross-pair variation, and any per-occupation version of it is reported as a distribution rather than as the estimate.

Two properties of the regressors split the test. Similarity is symmetric, $s_{ij} = s_{ji}$, and the wage differential is antisymmetric, since $\ln(w_j / w_i) = -\ln(w_i / w_j)$. Adding a pair effect μ_{ij} , symmetric by construction, to the equation absorbs similarity entirely, and with it every symmetric friction between the pair, and leaves the wage differential identified from the asymmetry of the pair's traffic alone: how many more people go from i to j than from j to i . The first specification, with origin and destination effects, therefore answers what similarity explains, which is the level of traffic between a pair; the second, with pair effects, answers what the wage differential explains, which is its direction. The flow account gives the reason to expect the two answers to differ. Across the whole matrix, gross lateral movement runs at 19.0 million people a year in 2022 to 2025 at the detailed level, of which 42.1 per cent carries direction at the pair level and 10.4 per cent changes the size of any occupation; over 2009 to 2025 the pair share is 32.3 per cent. Most of what a similarity score would predict is two-way traffic between neighbours, and what a workforce board wants to know, whether a move will fill a vacancy that stays filled, is a question about the smaller antisymmetric part.

The thresholds are tested by relaxing the functional form. In place of $\eta \ln s_{ij}$ the model takes a set of indicators for bins of similarity, of width 0.005 from 0.90 to 1 with everything below 0.90 as the base, and the sequence of bin coefficients is the curve of flow against similarity net of size and wages. A threshold is confirmed where the curve steps by more than the standard errors of the adjacent bins, and the three cutoffs in use, 0.985, 0.970 and 0.950, are placed against the steps the data contain. The

alternative reading, a smooth curve with no step anywhere, says that the tiers are a convenience of presentation and carry no information about where a move becomes hard.

4 What replaces licensing

The specification that preceded this one entered a licensing indicator at the destination as a regressor. It is dropped, for a reason the migration paper established: the survey's only credentialling variable measures any active certification or licence and is an upper bound on legal licensing, the compact timelines that would identify a licensing effect exist for one occupation, and a regressor that measures the wrong thing precisely is worse than none.

What replaces it is the residual map. The fitted model predicts a flow for every pair from similarity, wages and the two size effects, and the pairs where the observed flow falls far below the prediction, at a rate indistinguishable from zero on stocks large enough to see one, are the pairs where something the score does not carry blocks the move. The map is reported in full, as the pairs sorted by the ratio of observed to predicted flow with the standard error of the observed cell, and it is read three ways. Whether the blocked pairs concentrate in destinations that the Bureau of Labor Statistics classifies as requiring a licence is the licensing test, on every occupation the matrix covers rather than on a nurse example. Whether they concentrate in pairs whose destinations are geographically concentrated is the location test, using the interstate migration measures of the companion paper. And the pairs that remain after both are the information gaps, the moves that are feasible, permitted and available and that people do not make, which is where a bridge programme has something to do.

5 What is fixed before the estimation runs

The inputs are four and each has a source. The transition matrix is the flow account's within-interview build from the 2009 to 2025 supplements, pooled with year effects, on the Census 2010 occupation codes. Similarity is the composite over the Network's importance ratings, keyed to the Standard Occupational Classification and carried to Census codes by the account's bridge, with the employment-weighted mean similarity taken where a Census code maps to several classification codes. Wages are the May 2024 Occupational Employment and Wage Statistics national means, carried to Census codes the same way. Employment enters only through the fixed effects.

The readings are fixed. The primary estimate is η from the first specification with standard errors clustered by pair, reported with β from the second. The tier test is the bin curve and the location of its steps against 0.985, 0.970 and 0.950. The residual map is the full list. The result that ends the paper is η within two standard errors of zero on the pooled matrix, which says that a similarity score carries no information about who moves and that advice built on one is built on nothing; the result that vindicates the practice is an η large enough that a move from the first tier to the third cuts the predicted flow by more than half, with steps in the bin curve at or near the cutoffs in use. Whichever it is, the estimated thresholds replace the assumed ones in the shortage arithmetic of the companion paper, which takes them as given until this one reports.

6 Conclusion

A similarity score is a hypothesis, the moves people make are the evidence, and the two have been kept apart by the absence of a transition matrix that measures moves rather than coding noise. The flow account supplies the matrix, the Network supplies the score, and a gravity model on public data supplies the elasticity that every skill-based recommendation currently implies without stating. The model, its inputs and its readings are fixed here, and the estimation follows the matrix.

References

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1. Working paper. The specification, the bins, the thresholds under test and the readings are written to `output/tables/prespecification.txt` by `src/prespecify.py`, and `src/gravity.py` estimates the model from the matrix and the similarity table once they are placed in `data/processed/`. Every figure quoted from the flow account is taken from that paper’s published number file.